English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the handwritten solution outline. Original calculations and unfinished parts are retained. Apparent source inconsistencies are identified in transcription notes.

Problem 1

(a) Casino profit

Let X_i be the casino's profit from bet i. Then

$$\begin{array}{c|cc} X_1 & +1 & -1 \\ \hline \mathbb{P} & 19/37 & 18/37 \end{array}, \qquad \mathbb{E}X_1 = \frac{1}{37}, \qquad DX_1 = 1 - \left(\frac{1}{37}\right)^2 = \frac{1368}{1369} \approx 1.$$

We seek n such that $\mathbb{P}(X_1 + \cdots + X_n \geq 1000) \geq 90\%$, or equivalently

$$\mathbb{P}\left(\frac{X_1 + \cdots + X_n - n/37}{\sqrt{(1368/1369)n}} \geq \frac{1000 - n/37}{\sqrt{(1368/1369)n}}\right) \geq 90\%.$$

By the central limit theorem, use the standard normal approximation and require

$$\Phi\left(\frac{1000 - n/37}{\sqrt{(1368/1369)n}}\right) \leq 10\%,$$

approximately

$$1000 - \frac{n}{37} \leq -1.29\sqrt{\frac{1368}{1369}n}, \qquad n \geq 47387.$$

For a fixed n,

$$\begin{aligned} \mathbb{P}(X_1 + \cdots + X_n \geq 0) &= \mathbb{P}\left(\frac{X_1 + \cdots + X_n - n/37}{\sqrt{(1368/1369)n}} \geq -\sqrt{\frac{n}{1368}}\right) \\ &\approx 1 - \Phi\left(-\sqrt{\frac{n}{1368}}\right). \end{aligned}$$

Here $\sqrt{n/1368}$ is fairly large (about 6 when $n = 47387$), so the probability of nonnegative profit is approximately 1 and that of a loss approximately 0.

(b) Two bets

Let Y_1, Z_1 be the player's net winnings in the two games:

$$\begin{array}{c|cc} Y_1 & +1 & -1 \\ \hline \mathbb{P} & 18/37 & 19/37 \end{array} \qquad \begin{array}{c|cc} Z_1 & +35 & -1 \\ \hline \mathbb{P} & 1/37 & 36/37 \end{array}.$$

Then $\mathbb{E}Y_1 = \mathbb{E}Z_1 = -1/37$ and

$$DY_1 = 1 - (1/37)^2 \approx 1.00.$$

The source writes

$$DZ_1 = \frac{35^2}{37} + \frac{1^2 \cdot 36}{37} - \left(\frac{1}{37}\right)^2 \approx 1225.97. \qquad \text{(Note 1)}$$

Both games have the same expectation, but the second is riskier.

(c) Spending BGN 5

Game 1. For five separate single-number bets,

$$\mathbb{E}[Z_1 + \cdots + Z_5] = -\frac{5}{37}, \qquad D(Z_1 + \cdots + Z_5) = 5DZ_1 \approx 6129.86. \qquad \text{(Note 1)}$$

Game 2. For five distinct numbers in one spin, write Z_1^* for the net winnings:

$$\begin{array}{c|cc} Z_1^* & +31 & -5 \\ \hline \mathbb{P} & 5/37 & 32/37 \end{array}, \qquad \mathbb{E}Z_1^* = -\frac{5}{37}, \qquad DZ_1^* \approx 151.47.$$

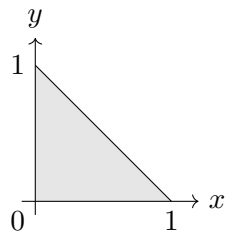
Game 3. For a BGN 5 red/black bet, write X_1^*:

$$\begin{array}{c|cc} X_1^* & +5 & -5 \\ \hline \mathbb{P} & 18/37 & 19/37 \end{array}, \qquad \mathbb{E}X_1^* = -\frac{5}{37}, \qquad DX_1^* \approx 24.98.$$

Problem 2

(a) The joint density

The support is the triangle $x \geq 0$, $y \geq 0$, $x + y \leq 1$.



Normalization gives

$$1 = \int_0^1 \int_0^{1-x} cx^3 y \, \mathrm{d}y \, \mathrm{d}x = c \int_0^1 x^3 \frac{(1-x)^2}{2} \, \mathrm{d}x = \frac{c}{120}, \qquad c = 120.$$

The marginal density is

$$f_X(x) = \int_0^{1-x} cx^3 y \, \mathrm{d}y = cx^3 \frac{(1-x)^2}{2}, \qquad x \in [0,1],$$

and

$$\begin{aligned} \mathbb{E}Y &= \int_0^1 \int_0^{1-x} y \, cx^3 y \, \mathrm{d}y \, \mathrm{d}x \\ &= c \int_0^1 x^3 \frac{(1-x)^3}{3} \, \mathrm{d}x = \frac{120}{420} = \frac{2}{7} \approx 0.2857. \end{aligned}$$

(b) Conditional expectations

Clearly, $\mathbb{E}[X \mid X = 1/2] = 1/2$. Also,

$$\mathbb{E}[Y \mid X = 1/2] = \int_0^{1/2} y \frac{c(1/2)^3 y}{c(1/2)^3(1/2)^2/2} \, \mathrm{d}y = 8 \int_0^{1/2} y^2 \, \mathrm{d}y = \frac{1}{3}.$$

(c) The product

Set $Z = XY$ and $W = X$. Then $X = W$, $Y = Z/W$ and

$$\det J = \begin{vmatrix} 0 & 1 \\ 1/w & -z/w^2 \end{vmatrix} = -\frac{1}{w}.$$

Consequently,

$$\begin{aligned} f_{Z,W}(z,w) &= f_{X,Y}(w, z/w) \frac{1}{|w|} \\ &= cw^3 \frac{z}{w} \frac{1}{w} \mathbf{1}_{\{w+z/w \leq 1, \ z,w \geq 0\}} = czw \mathbf{1}_{\{z,w \geq 0, \ z \leq w - w^2\}}. \end{aligned}$$

The condition is $w^2 - w + z \le 0$, whose roots are

$$w_{1,2} = \frac{1 \mp \sqrt{1-4z}}{2}.$$

Thus $0 \le z \le 1/4$ and w lies between these roots. Integrating,

$$\begin{aligned} f_Z(z) &= \int_{(1-\sqrt{1-4z})/2}^{(1+\sqrt{1-4z})/2} czw \, \mathrm{d}w \\ &= \frac{cz}{8}\left[(1+\sqrt{1-4z})^2 - (1-\sqrt{1-4z})^2\right] \\ &= \frac{cz}{8} 4\sqrt{1-4z} = \frac{cz\sqrt{1-4z}}{2}, \qquad z \in [0, 1/4]. \end{aligned}$$

Problem 3

(a) Red cells

Let X be the number of red cells, so $X \sim \mathrm{Bin}(81, 1/3)$. Then

$$\mathbb{P}(X \le 21) = \mathbb{P}\left(\frac{X - 81/3}{\sqrt{81 \cdot \frac{1}{3} \cdot \frac{2}{3}}} \le \frac{21 - 81/3}{\sqrt{81 \cdot \frac{2}{9}}}\right) \approx \Phi(-1.41) \approx 7.93\%,$$

using the central limit theorem (why is it applicable?).

(b) Flags

No: possible flag positions overlap. Think about a formal proof!

(c) Expected number of flags

The possible positions are 7 per row times 9 rows. Number them. If X_i is the indicator that position i contains a flag, then $X_i \sim \mathrm{Ber}(1/27)$. The source writes

$$\mathbb{E}[\text{number of vertical flags}] = \mathbb{E}[X_1 + \cdots + X_{67}] = 67\mathbb{E}X_1 = \frac{67}{27}. \qquad \text{(Note 2)}$$

(d) Both orientations

They are not independent (why?). By linearity, the original concludes

$$\mathbb{E}[\text{vertical} + \text{horizontal}] = 2\mathbb{E}[\text{vertical}] = \frac{134}{27}. \qquad \text{(Note 2)}$$

Problem 4

(a) Sums of exponentials

By convolution,

$$f_{X+Y}(x) = \int_{-\infty}^{\infty} f_X(y) f_Y(x-y) \, \mathrm{d}y.$$

Hence, for $x \ge 0$,

$$f_{S_2}(x) = \int_{-\infty}^{\infty} e^{-y} e^{-(x-y)} \mathbf{1}_{\{y, x-y \ge 0\}} \, \mathrm{d}y = \int_0^x e^{-x} \, \mathrm{d}y = xe^{-x}.$$

For the induction, the source labels the preceding density $f_{S_{n-1}}(x) = x^{n-1}e^{-x}\mathbf{1}_{\{x>0\}}/(n-1)!$ and writes

$$\begin{aligned} f_{S_n}(x) &= \int_{-\infty}^{\infty} \frac{y^{n-1}}{(n-1)!} e^{-y} e^{y-x} \mathbf{1}_{\{y, x-y \ge 0\}} \, \mathrm{d}y \\ &= e^{-x} \int_0^x \frac{y^{n-1}}{(n-1)!} \, \mathrm{d}y = \frac{x^n}{n!} e^{-x} \mathbf{1}_{\{x>0\}}. \qquad \text{(Note 3)} \end{aligned}$$

(b)

…

(c) The counting variable

$$\begin{aligned}\mathbb{P}(N_t = n) &= \mathbb{P}(S_n \le t) - \mathbb{P}(S_{n+1} \le t) \\ &= \int_0^t \frac{x^{n-1}}{(n-1)!} e^{-x}\,\mathrm{d}x - \int_0^t \frac{x^n}{n!} e^{-x}\,\mathrm{d}x.\end{aligned}$$

Integration by parts gives

$$\int_0^t \frac{x^n}{n!} e^{-x}\,\mathrm{d}x = \left[-\frac{x^n}{n!} e^{-x}\right]_0^t + \int_0^t e^{-x} \frac{x^{n-1}}{(n-1)!}\,\mathrm{d}x.$$

The repeated integrals cancel, yielding

$$\mathbb{P}(N_t = n) = e^{-t} \frac{t^n}{n!}.$$

The source's final boxed statement is $N_t \sim \mathrm{Poi}(1)$. See Note 4.

Transcription notes

1. The variance expression for Z_1 is approximately 34.08, not 1225.97. The source carries its stated value into the five-bet variance as well.
2. The source states $7 \cdot 9$ positions but uses 67 in the subsequent count. There are 63 positions for either orientation, so the displayed expectations require review.
3. The induction labels are shifted: the density $x^{n-1}e^{-x}/(n-1)!$ belongs to S_n and its convolution with an additional exponential belongs to S_{n+1}.
4. The displayed probability mass function has Poisson parameter t. The source's final parameter 1 is inconsistent with that mass function.